CANADIAN BIOINFORMATICS WORKSHOP

Get ready for LLMs in Bioinformatics

A short, practical setup guide for students

NOVEMBER 14-15, 2026

10 minutes required / local setup optional

START HERE

A light-touch pre-work check

1	**Laptop + charger**	Bring a laptop you can use in a browser.
2	**Google Colab access**	Sign in and run one tiny Python cell.
3	**Slides + notebook**	Open the module slides beside its linked notebook.
4	**Questions are welcome**	If setup fails, bring the error; a TA can help.

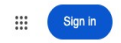
Not required for this pre-work

- Paid plans
- GPU access
- GitHub or Copilot account
- A pre-installed package bundle

If the final lab instructions change a requirement, the organizers will say so explicitly.

REQUIRED · ABOUT 10 MINUTES

Open Colab and test one cell



Colab landing page, signed out. Captured Nov 9, 2026.

Open colab.research.google.com and sign in with a Google account.

Run this in a code cell

```
print("Hello, bioinformatics!")
```

Your check is complete when the output appears below the cell.

Save a copy of a workshop notebook before editing. Colab sessions are temporary; save work you want to keep.

DURING THE WORKSHOP

Let the slides and notebook work as a pair

01

Slides

Frame the biological question and the method.

02

Notebook

Open the linked lab, save a Drive copy, follow setup notes.

03

Run + inspect

Execute cells in order. Read the output; do not just race to the end.

04

Return to slides

Use the checkpoint to connect results back to the concept.

Your instructor will share the notebook link for each lab before the activity.

ONLY IF YOU WANT TO PREPARE

Local Python setup is optional



Official VS Code Python tutorial, captured Nov 9, 2026.

Use the current vendor instructions if the page changes.

What you need

1. Python 3 interpreter
2. VS Code editor
3. Microsoft Python extension
4. Optional: Jupyter extension for local notebooks

```
print("Hello, bioinformatics!")
```

Do not install workshop libraries until exact lab requirements are confirmed.

Ready is enough.

Bring your laptop, your questions, and your curiosity.

Keep research data safe

Use public or synthetic workshop data only. Never upload identifiable, clinical, confidential, unpublished, or restricted data to personal cloud notebooks or AI tools.

If setup fails, bring the error message with private details removed. A TA can help.

See the student guide for official links and step-by-step instructions.

NOT NEEDED FOR THIS WORKSHOP

Optional AI access programs

GitHub Copilot Student

For verified students in eligible degree- or diploma-granting programs. Apply through GitHub Education with current enrollment proof, then activate Copilot Student separately after approval.

Start: github.com/settings/education/benefits

See the guide for official application instructions.

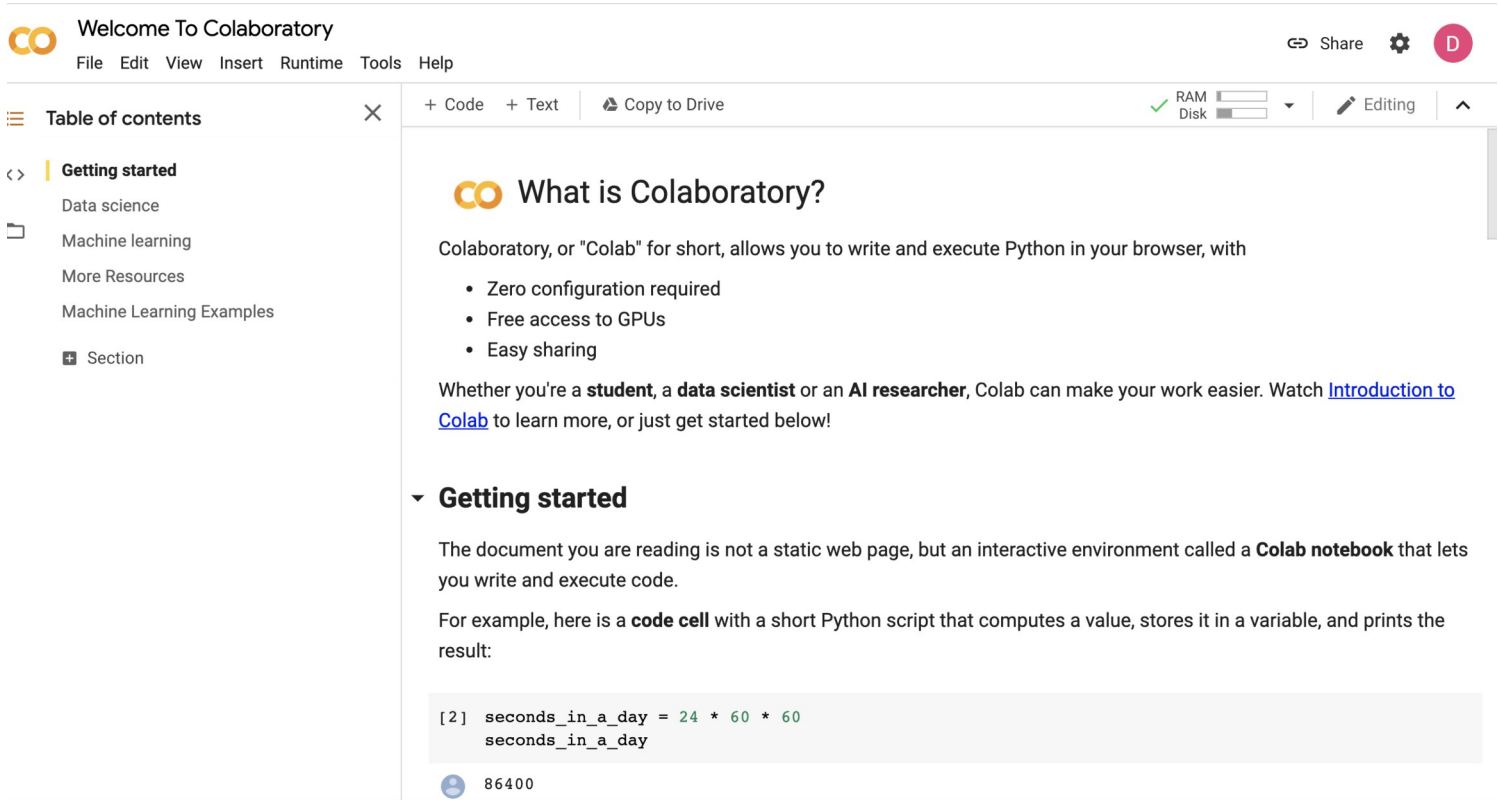
Claude for research labs

Anthropic's scientist plan is PI-led: an eligible academic/nonprofit PI verifies first, then may add lab members. Announced standard seats are free and premium seats are US$15/month for one year; confirm current availability and terms.

Students: ask your PI; workshop registration alone does not qualify you.

No paid AI account is required for pre-work.

Inside a Colab notebook



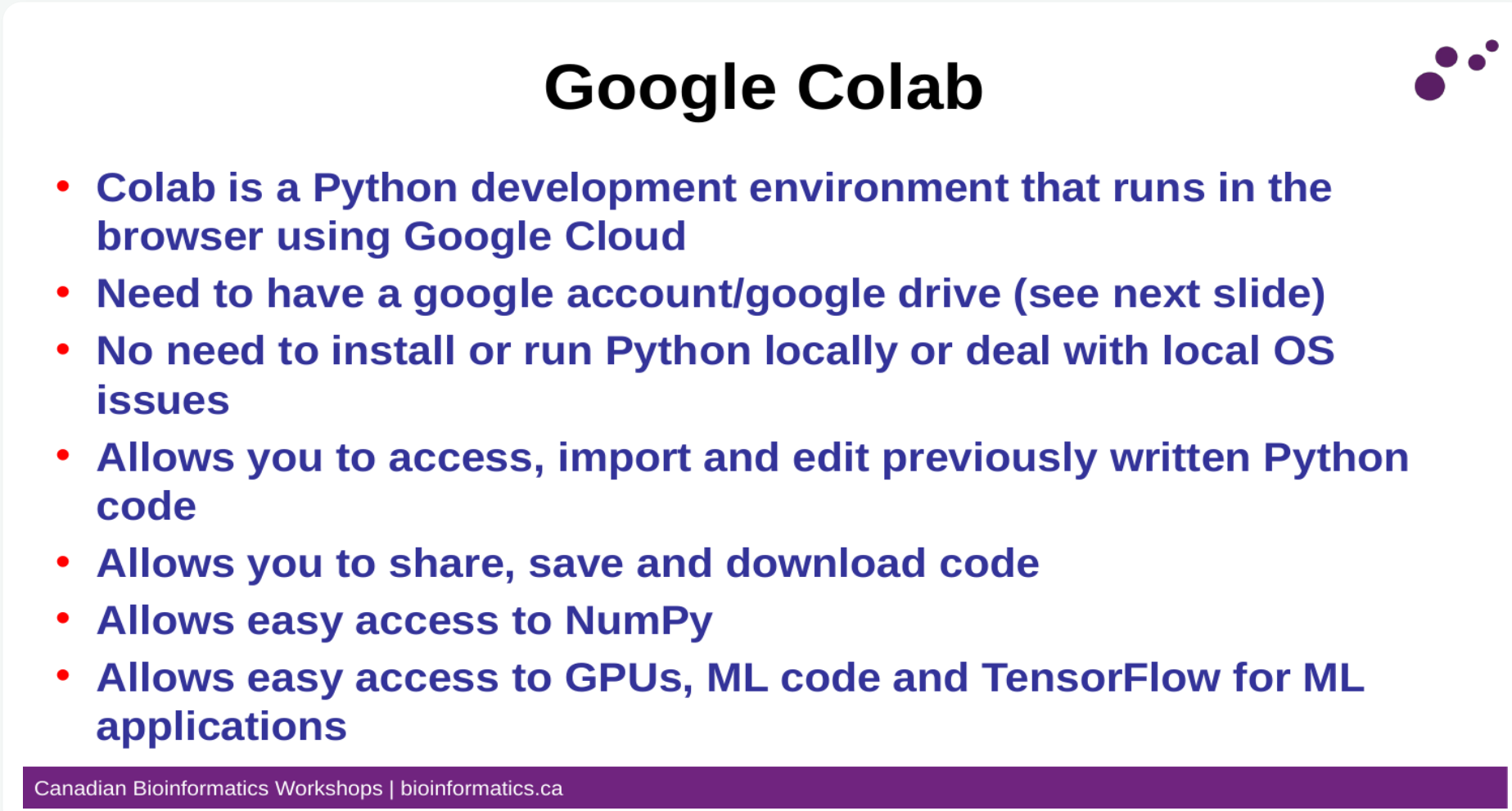
A notebook has

Code cells
Outputs beneath cells
Text and explanation
Controls to add cells and save a copy

Historical screenshot reused from Module 1 of the April 2026 Machine Learning workshop. Current Colab screens may differ.

Use the current official Colab page and the workshop-provided notebook link.

Google Colab introduction

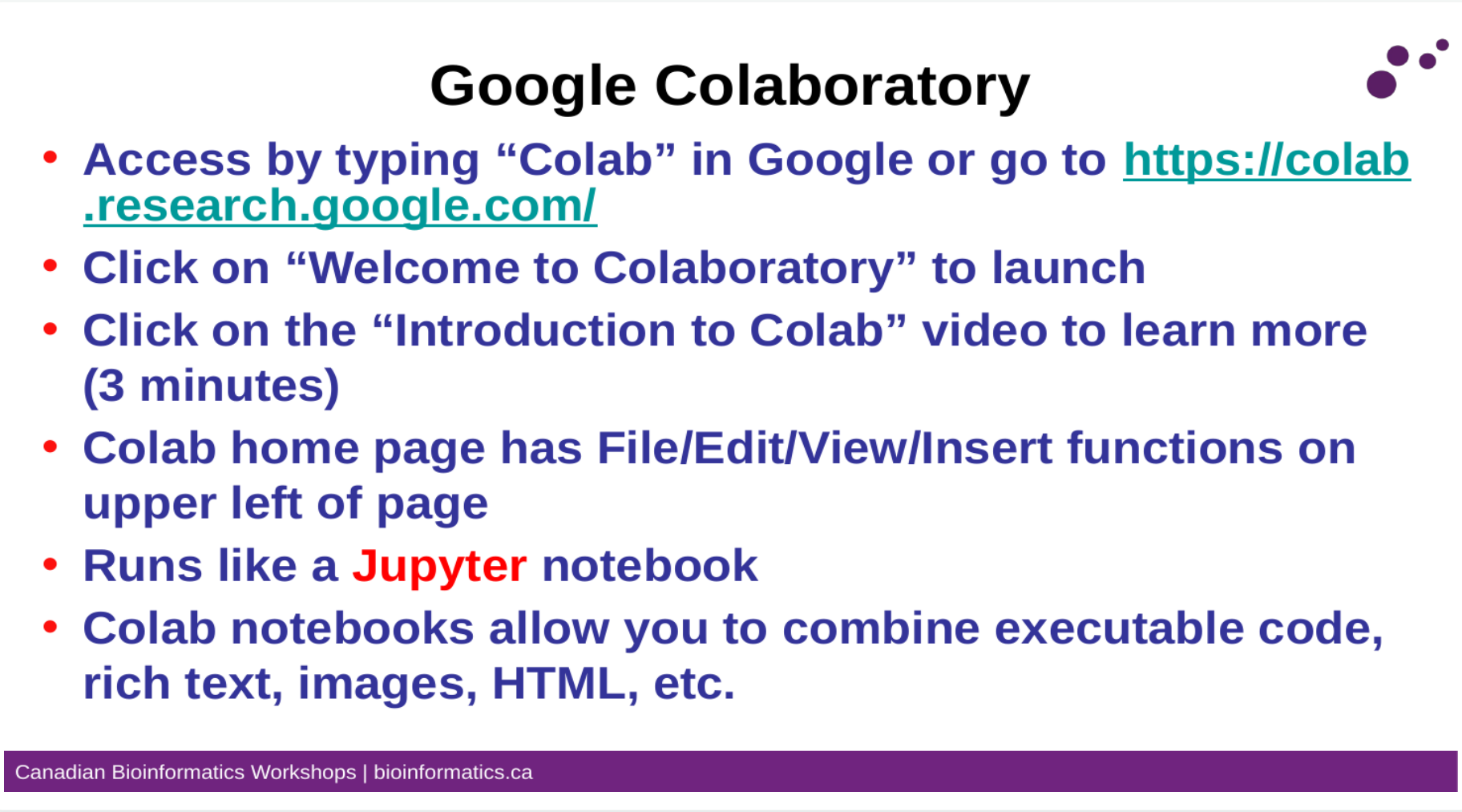


Historical example

MLE 0426
Module 1
Slide 090
April 2026

This is a historical interface walkthrough. Colab screens and service features can change; follow current official instructions.

Colab access



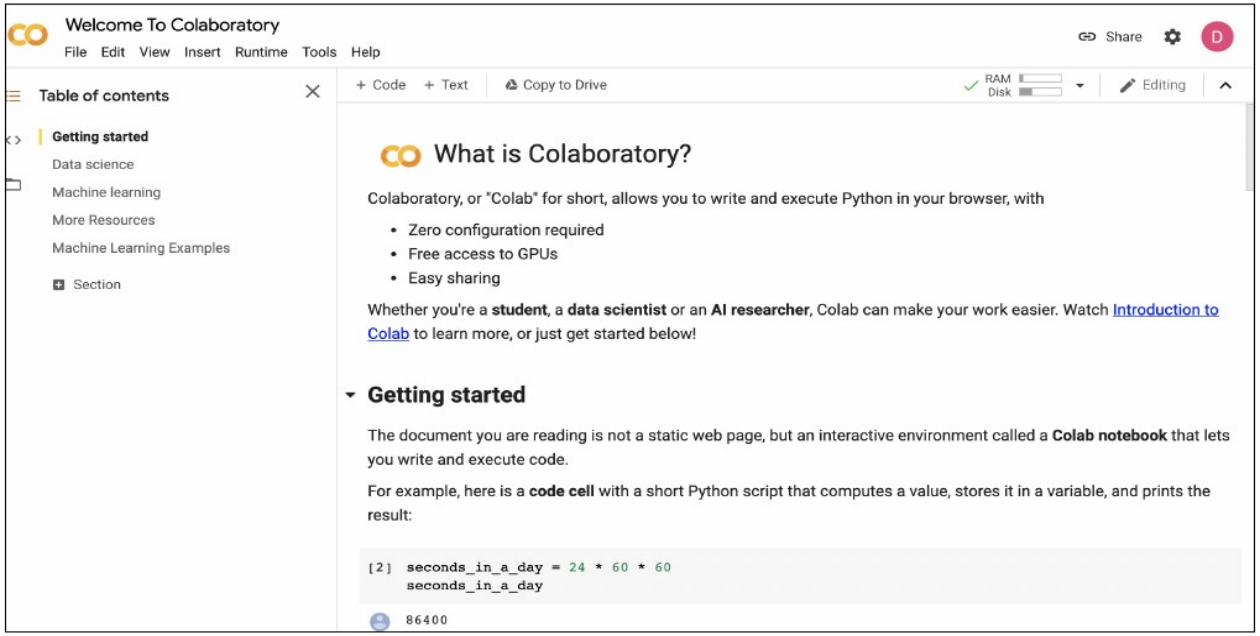
Historical example

MLE 0426
Module 1
Slide 097
April 2026

This is a historical interface walkthrough. Colab screens and service features can change; follow current official instructions.

Colab notebook interface

Colab Main Page



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Historical example

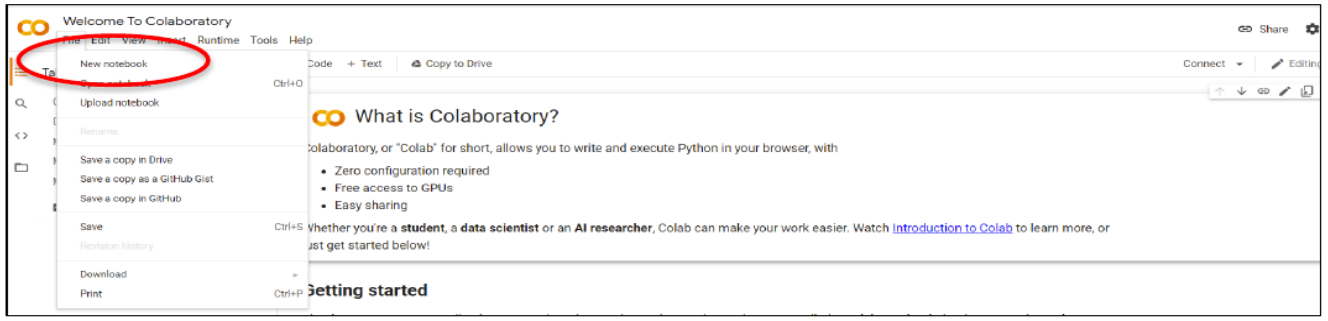
MLE 0426
Module 1
Slide 108
April 2026

This is a historical interface walkthrough. Colab screens and service features can change; follow current official instructions.

Creating a notebook

How to Open a New Notebook

- Step 2: Under File, click New Notebook
- A new notebook should open up as shown below





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Historical example

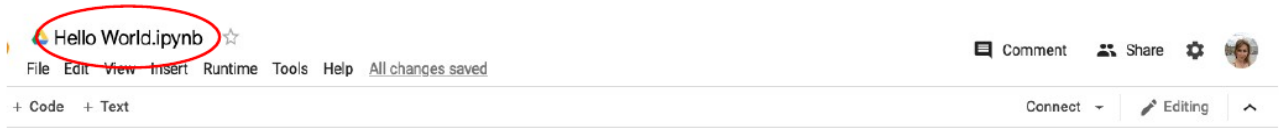
MLE 0426
Module 1
Slide 111
April 2026

This is a historical interface walkthrough. Colab screens and service features can change; follow current official instructions.

Hello World example

Hello World

- Let’s write a program in Colab and call it “Hello World”
- Type that name in the circle and press Enter
- The notebook will acquire the name you have given now



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Historical example

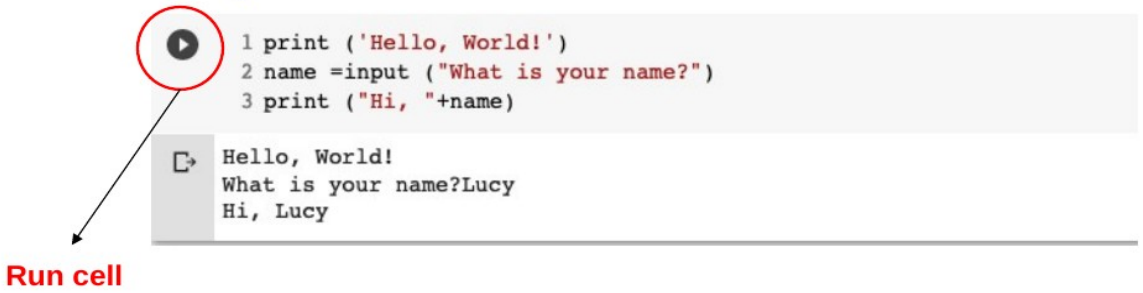
MLE 0426
Module 1
Slide 115
April 2026

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Running a code cell

Writing and Executing the Code

- Enter the trivial Python code and run it
- We will show how the simple code “Hello World” can be written
- Write this few lines of code in code cell
- Press run cell
- The output are the results



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Historical example

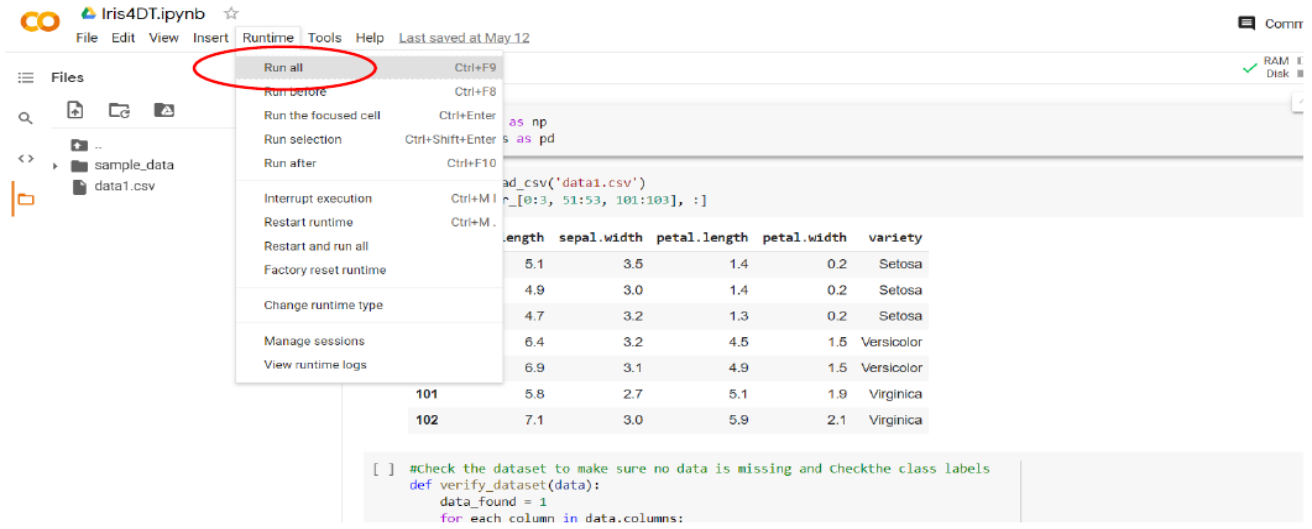
MLE 0426
Module 1
Slide 116
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Running all cells

Runtime – Run All

- To save time, you can also click ‘Runtime’ then select ‘Run all’ to run all the cells



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Historical example

MLE 0426
Module 1
Slide 125
April 2026

This is a historical interface walkthrough. Colab screens and service features can change; follow current official instructions.